



From Propositional to Perceptual Asymmetry: Extending Frictive Policy Optimization to Asymmetric Partial Information Dialogue

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Motivation

A Communication Breakdown in interaction when participants having different information.

The **difference in perception** results in a **difference in understanding**.



AI-Generated image

Motivation



AI-Generated images



Research Question

When two people are working from different information, **can LLMs spot their misunderstandings better by seeing only one person's view, or by seeing both?**

- Finding: Restricting the model to a single participant's view beats, in all three models we tested.
- Contribution: We show how to evaluate friction from inside one participant's view and that the same signal predicts the repairs the dialogue actually makes.

A short horizontal bar with a teal segment on the left and an orange segment on the right.

Outline of the talk

- Dataset: where the data comes from, and what counts as a misunderstanding.
- Probing methodology: how we test whether perspective helps.
- Results and discussion: what we found, and what follows from it.



Outline of the talk

- **Problem we found:** conversation breakdown caused by perceptive-based asymmetry
- **Solution we proposed:** introduce perspective-based strategies for friction detection and repair
- **Does the solution help in the friction detection and repair predictions in LLM?**
 - Yes, limiting the perspective helps the performance in friction detection and repairs.

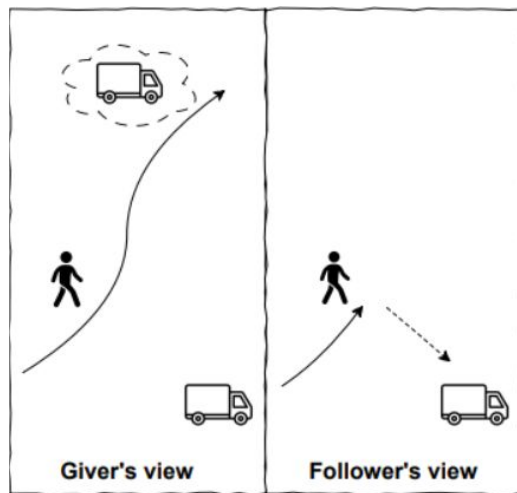


Dataset - Maptask^[1]

- 2 participants, 2 maps, different landmarks
 - Giver: has a route marked on the giver's map and must guide the other participant
 - Follower: reproduce this route on the follower's map through verbal communication alone.
- Why this breakdown happened?
 - Because participants have different information.

Giver: Go to the parked van ◦ ◦

the upper
van



Follower: ok, to the **bottom**. ◦ ◦ ◦

The
bottom
van

Giver: NO! ◦ ◦ ◦

The bottom
van != The
upper van

[1] Anne H. Anderson, Miles Bader, Ellen Gurman Bard, Elizabeth Boyle, Gwyneth Doherty, Simon Garrod, Stephen Isard, Jacqueline Kowtko, Jan McAllister, Jim Miller, Catherine Sotillo, Henry S. Thompson, and Regina Weinert. 1991. The HCRC Map Task corpus. *Language and Speech*, 34(4):351–366



What's the Problem?

- Each participant perceives different evidence
 - each constructs different propositional content from what they perceive
 - their belief states diverge
 - that divergence surfaces as ambiguity, misreference, and misunderstanding in the dialogue.
- The giver's evidence set contains two vans; the follower's contains one. Same words, different grounding, because different evidence.



What's the Extension?

- Work on common ground and common-ground tracking with LLMs (e.g., DeliData, Karadzhov et al., 2023) , including our own (WTD, Khebour et al., 2024), assumes the evidence is spoken or written text, shared by all participants.
- Two different Asymmetries
 - **Propositional asymmetry:** participants receive the ***same* information** and draw different conclusions from it. Addressed in prior work.
 - **Perceptual asymmetry:** participants have private perceptual access, so their ***evidence itself* differs**. Not addressed in prior work.
- We evaluate friction from within each participant's own perspective rather than over a shared record.

Ibrahim Khalil Khebour, Mariah Bradford, Kenneth Lai, Christopher Tam, Jingxuan Tu, Benjamin A.Ibarra, Nathaniel Blanchard, Nikhil Krishnaswamy, and James Pustejovsky. 2024. Common ground tracking in multimodal dialogue. In Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024), pages 3587–3602.

Georgi Karadzhov, Tom Stafford, and Andreas Vlachos. 2023. DeliData: A dataset for deliberation in multiparty problem solving. Proceedings of the ACM on Human-Computer Interaction, 7(CSCW2):1–25.



Two different Asymmetries

- Previous common ground tracking work, eg. Weights Task (Khebour et al., 2024), DeliData (Karadzhov et al., 2023) assumes the interaction of spoken and written texts
 - same perception, different interpretation (**propositional asymmetry**)
- But this is not the case for this scenario:
 - Different evidence -> different interpretation (**perceptual Asymmetry**)-> epistemic misalignment (friction)
- Enrich the policy optimization to actually take perception into considerations, arrival of steady state of the common ground

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Perspective-Based Evaluation

- Show the problem is real and structured
 - (MapTask corpus): perceptual asymmetry is common in dialogue and interaction and regular and pattern-based
- Propose a solution
 - A perspective-based friction detection, from each participant's interpretation of the current situation
- Does the solution work?
 - A perspective-based friction detection, from each participant's interpretation of the current situation



Problem is real

Among three understanding state: aligned, pending, misunderstood (Li et al. 2025):

- Multiplicity, which is 7% of corpus, but 12 % of the misunderstanding
- 38% of pending is ungrounded — follower can't commit to a candidate, exactly the Uncertainty diagnostic.



Corpus Evidence

HCRC MapTask, 13,077 reference expressions across 128 dialogues, annotated per expression for the giver's *intended* and the follower's *interpreted* landmark. Each expression is aligned, pending, or misunderstood.

- Finding one, **multiplicity**: The parked-van example is a *multiplicity* case: one participant's map has two instances of a landmark, the other's has one. Multiplicity is 7% of reference expressions but produces 12% of misunderstandings, six times the corpus average.
- Finding two, **pending**: Misalignment surfaces in two places: outright misunderstanding, and *pending*, where grounding never completes. 38% of pending cases are multiplicity.

Other discrepancy types exist in the corpus and produce other friction, but multiplicity is the dominant source; the rest of the talk focuses on it.



The Perspective-Based friction detection

4 failure modes, corresponding to perspective-based diagnostic (Pustejovsky & Krishnaswamy 2026)

Value conflict - Is the expression I am producing (or accepting) uniquely identifying for me, and if not, have I flagged it?

Hazard - Am I (or my partner) acting confidently on a grounding that I have reason to doubt?

Contradiction - Given the spatial claims I attribute to my partner, is there a candidate on my map that satisfies both?

Uncertainty - Can I commit to a referent, or does my evidence admit multiple candidates?



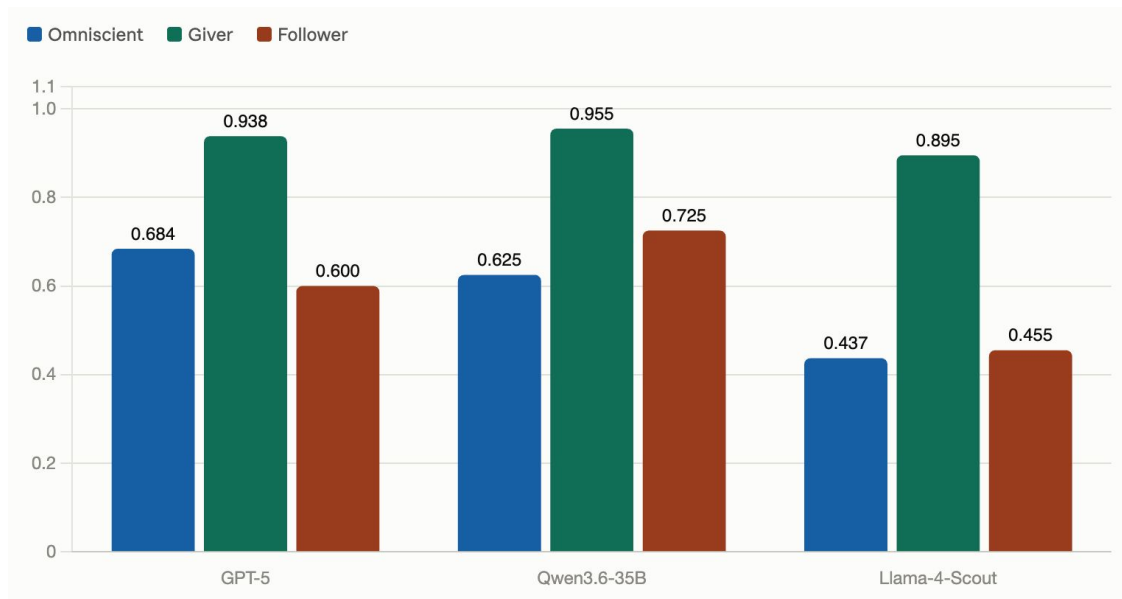
Probe1: Does the LLM understand the perspective differences?

- Omniscient: Both maps + dialogue history
 - Question: Do giver and follower refer to the same physical landmark?
- Giver-only: Giver's map only + dialogue history
 - Question: is the target RE underspecified on the giver's own map?
- Follower-only: Follower's map only + dialogue history
 - Question: does the giver's spatial description conflict with the follower's map?



More information makes it worse

Consistent across closed/open, dense/MoE, full/quantized models — the gap is driven entirely by the friction subset, not the easy controls.





Probe2: Is the perspectival signal not only detectable but productive?

- Omniscient: Both maps + dialogue history
 - Target: contradiction detected over a global perspective
 - $\text{Contr}(h, y)$
- Perspective: Giver's map only + dialogue history vs. follower's map only + dialogue history
 - Target: contradiction detected between two perspectives
 - $\text{Contr}(h^{(i)}, y) = \mathbf{1}[\text{ref}(y \mid h^{(g)}) \neq \text{ref}(y \mid h^{(f)})]$



Help in Friction detection and Predicting Real Repairs

	Gemma-4-31B		GPT-5	
	Omni.	Persp.	Omni.	Persp.
Repairs (of 13)	5	13	12	13

- Real repair:
after misunderstanding,
shared belief on the RE
within the 5 REs in the real
dialogue



Summary & Future Work

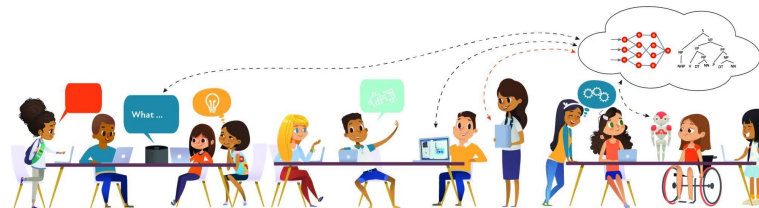
- Collaborative dialogue is not always a **shared-context** phenomenon: when participants hold private perceptual state, friction lives in the gap between their information horizons.
- **Perspectival re-annotation:** Re-annotate OneCommon by applying the grounding cascade separately to each participant's visual perspective, testing whether friction diagnostics generalize to continuously varying asymmetry.
- **Cross-task transfer:** Evaluate the perspective-based patterns on other structurally asymmetric dialogue tasks, including DPIP (Zhu et al., 2026).
- **Epistemic alignment:** Develop perspectival annotations that distinguish surface coordination from genuine belief alignment, revealing silent accommodation and persistent interpretive divergence.



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Beyond MapTask

OneCommon (Udagawa & Aizawa, 2019): continuous, not discrete, perceptual overlap — a much harder setting.

- 56% selection-failure rate (vs. 1.8% in MapTask) — but the same principle holds.
- Quantity of friction doesn't predict success: failed dialogues hedge more